Definitions

Definition (Independent Set)

Let $G = (V, E)$ be a graph. An *independent set* in G is a subset $S \subseteq V$ such that there are no edges entirely in S. That is, $\{u, v\} \in E$ implies that $u \notin S$ or $v \notin S$.

Definitions

Definition (Independent Set)

Let $G = (V, E)$ be a graph. An *independent set* in G is a subset $S \subseteq V$ such that there are no edges entirely in S. That is, $\{u, v\} \in E$ implies that $u \notin S$ or $v \notin S$.

Definition (Coloring)

For an undirected graph $G = (V, E)$, a *k-coloring* of G is a mapping $f : V \to [k]$ such that for all edges $\{u, v\} \in E$, we have $f(u) \neq f(v)$.

Definitions

Definition (Independent Set)

Let $G = (V, E)$ be a graph. An *independent set* in G is a subset $S \subseteq V$ such that there are no edges entirely in S. That is, $\{u, v\} \in E$ implies that $u \notin S$ or $v \notin S$.

Definition (Coloring)

For an undirected graph $G = (V, E)$, a *k-coloring* of G is a mapping $f : V \to [k]$ such that for all edges $\{u, v\} \in E$, we have $f(u) \neq f(v)$.

Graph k-Coloring problem: Given a graph $G = (V, E)$ and a natural number k, output a k-coloring of G (if one exists).

The Result

- **2-coloring:** (Your SRE 2 today) BFS solves it in $O(n + m)$ time.

The Result

- **2-coloring:** (Your SRE 2 today) BFS solves it in $O(n + m)$ time.
- **3-coloring**

The Result

- **2-coloring:** (Your SRE 2 today) BFS solves it in $O(n + m)$ time.
- **3-coloring** Exhaustive search - try 3^n assignments, checking every edge. Time: $O((n + m)3^n)$.

The Result

- **2-coloring:** (Your SRE 2 today) BFS solves it in $O(n + m)$ time.
- **3-coloring** Exhaustive search - try 3^n assignments, checking every edge. Time: $O((n + m)3^n)$.

Theorem. 3-coloring can be solved in time

$$O(1.89^n).$$

Still exponential, but with a smaller base.

The Result

- **2-coloring:** (Your SRE 2 today) BFS solves it in $O(n + m)$ time.
- **3-coloring** Exhaustive search - try 3^n assignments, checking every edge. Time: $O((n + m)3^n)$.

Theorem. 3-coloring can be solved in time

$$O(1.89^n).$$

Still exponential, but with a smaller base.
Crucial to this theorem is the connection between Coloring and Independent Set.

Coloring and Independent Set

Color classes are independent sets.
A 3-coloring partitions V into three independent sets C_0, C_1, C_2, one for each color. Some classes may be empty.

Coloring and Independent Set

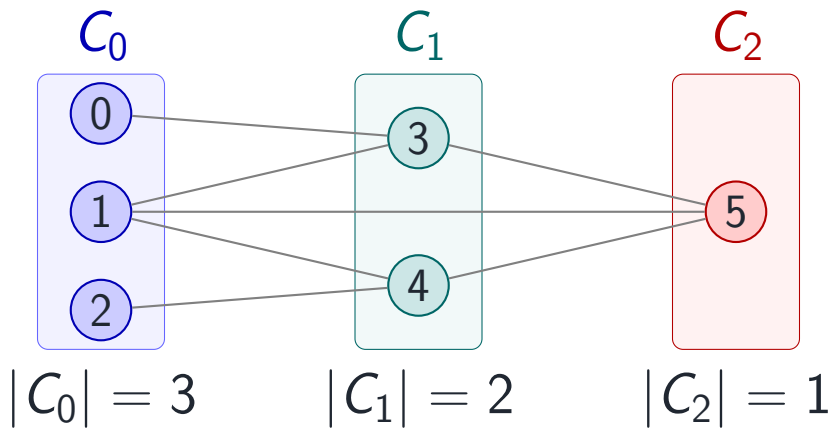
Color classes are independent sets.
A 3-coloring partitions V into three independent sets C_0, C_1, C_2, one for each color. Some classes may be empty.

$$|C_0| + |C_1| + |C_2| = n \quad \Longrightarrow \quad \min\{|C_0|, |C_1|, |C_2|\} \leq \left\lfloor \frac{n}{3} \right\rfloor .$$

Coloring and Independent Set

Color classes are independent sets.
A 3-coloring partitions V into three independent sets C_0, C_1, C_2, one for each color. Some classes may be empty.

$$|C_0| + |C_1| + |C_2| = n \quad \Longrightarrow \quad \min\{|C_0|, |C_1|, |C_2|\} \le \left\lfloor \frac{n}{3} \right\rfloor .$$



The Proof Idea

Idea: guess only the smallest color class S.

- Check that S is independent.
- Remove S and test whether the remaining graph is 2-colorable.

The Proof Idea

Idea: guess only the smallest color class S.

- Check that S is independent.
- Remove S and test whether the remaining graph is 2-colorable.

$$G_{\setminus S} = (V \setminus S, E_{\setminus S}), \qquad E_{\setminus S} = \{\{u, v\} \in E : u, v \notin S\}.$$

The Proof Idea

Idea: guess only the smallest color class S.

- Check that S is independent.
- Remove S and test whether the remaining graph is 2-colorable.

$$G_{\setminus S} = (V \setminus S, E_{\setminus S}), \qquad E_{\setminus S} = \{\{u, v\} \in E : u, v \notin S\}.$$

Lemma

1. If G is 3-colorable, some independent set S with $|S| \leq \lfloor n/3 \rfloor$ leaves $G_{\setminus S}$ 2-colorable.

The Proof Idea

Idea: guess only the smallest color class S.

- Check that S is independent.
- Remove S and test whether the remaining graph is 2-colorable.

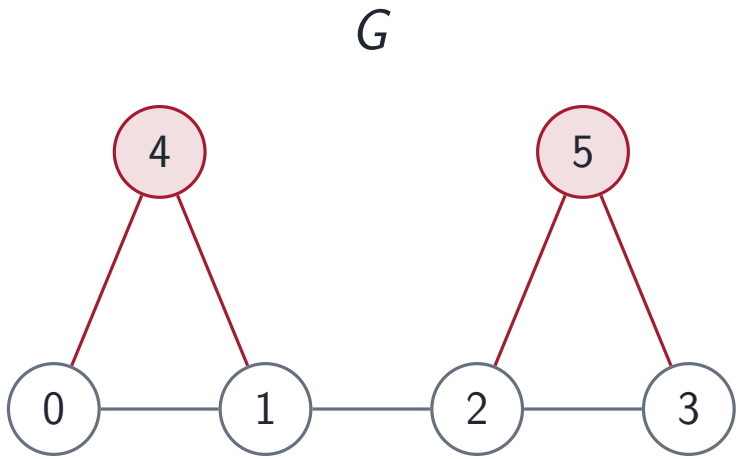
$$G_{\setminus S} = (V \setminus S, E_{\setminus S}), \qquad E_{\setminus S} = \{\{u, v\} \in E : u, v \notin S\}.$$

Lemma

1. If G is 3-colorable, some independent set S with $|S| \leq \lfloor n/3 \rfloor$ leaves $G_{\setminus S}$ 2-colorable.
2. If S is independent and $G_{\setminus S}$ is 2-colorable, then G is 3-colorable.

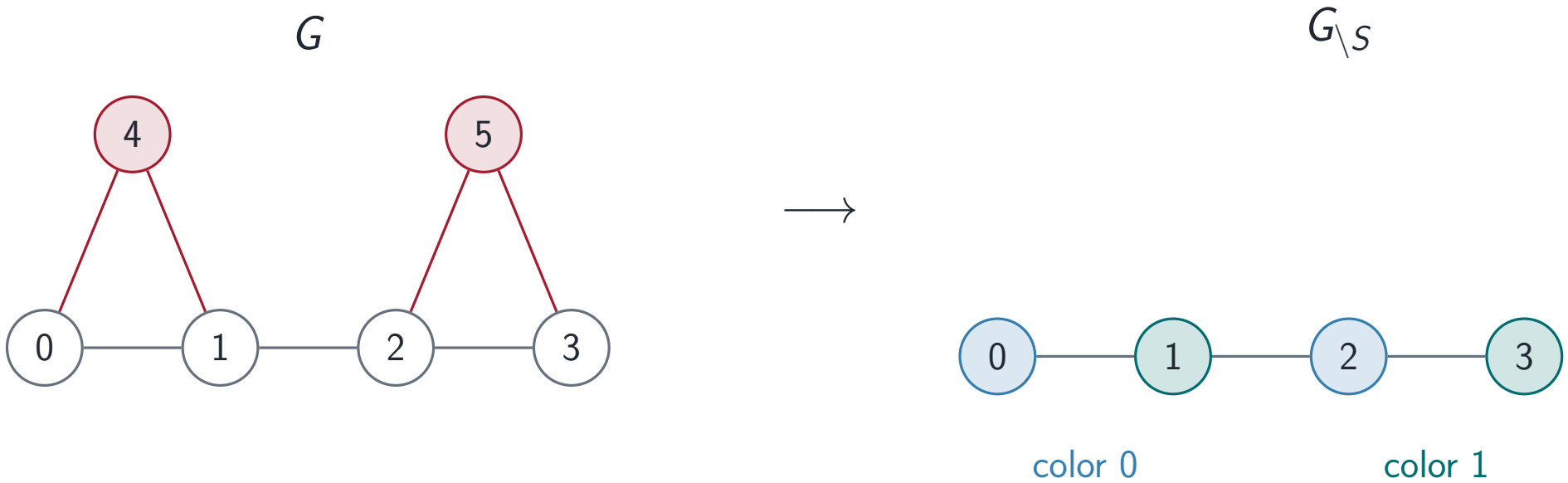
The Proof

Example: $n = 6$ **and** $S = \{4, 5\}$**.** The set S is independent and $|S| = n/3$.



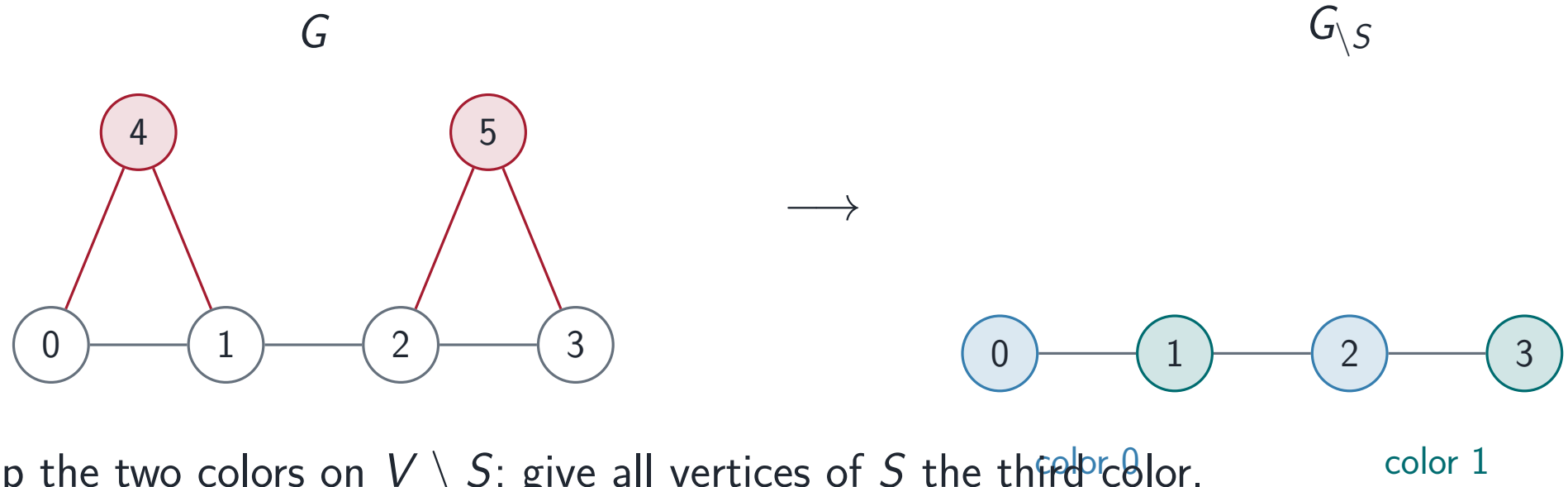
The Proof

Example: $n = 6$ **and** $S = \{4, 5\}$**.** The set S is independent and $|S| = n/3$.



The Proof

Example: $n = 6$ **and** $S = \{4, 5\}$**.** The set S is independent and $|S| = n/3$.



Keep the two colors on $V \setminus S$; give all vertices of S the third color.

The Proof

Proof of the lemma

- **Part 1:** take the smallest color class as S. The other two classes give a 2-coloring of $G_{\setminus S}$.

The Proof

Proof of the lemma

- **Part 1:** take the smallest color class as S. The other two classes give a 2-coloring of $G_{\setminus S}$.
- **Part 2:** let $g : V \setminus S \to \{0, 1\}$ be a 2-coloring. Define

$$f(v) = \begin{cases} g(v), & v \notin S, \\ 2, & v \in S. \end{cases}$$

The Proof

Proof of the lemma

- **Part 1:** take the smallest color class as S. The other two classes give a 2-coloring of $G_{\setminus S}$.
- **Part 2:** let $g : V \setminus S \to \{0, 1\}$ be a 2-coloring. Define

$$f(v) = \begin{cases} g(v), & v \notin S, \\ 2, & v \in S. \end{cases}$$

Edges outside S are properly colored by g. Edges crossing S get color 2 at exactly one endpoint. There are no edges inside S.

The Proof: Algorithm

Input: a graph $G = (V, E)$. **Output:** a 3-coloring, or $\perp$.

```
1 foreach S ⊆ V with |S| ≤ ⌊n/3⌋ do
2     if S is independent then
3         Construct G_{\S}
4         g ← BFSColoring(G_{\S})
5         if g ≠ ⊥ then
6             Set f(v) ← g(v) for v ∉ S, and f(v) ← 2 for v ∈ S
7             return f
8 return ⊥
```

The Proof: Algorithm

Input: a graph $G = (V, E)$. **Output:** a 3-coloring, or $\perp$.

```
1 foreach S ⊆ V with |S| ≤ ⌊n/3⌋ do
2     if S is independent then
3         Construct G_{\S}
4         g ← BFSColoring(G_{\S})
5         if g ≠ ⊥ then
6             Set f(v) ← g(v) for v ∉ S, and f(v) ← 2 for v ∈ S
7             return f
8 return ⊥
```

Include $S = \varnothing$. Run the 2-coloring test in *every* connected component.

The Proof: Runtime

Work for each candidate S: $O(n+m)$.

- Check that S is Independent Set and construct $G_{\setminus S}$.
- Run BFS to 2-color the remaining graph; construct the output if successful.

The Proof: Runtime

Work for each candidate S: $O(n+m)$.

- Check that S is Independent Set and construct $G_{\setminus S}$.
- Run BFS to 2-color the remaining graph; construct the output if successful.

Number of candidates:

$$N(n) = \sum_{k=0}^{\lfloor n/3 \rfloor} \binom{n}{k}.$$

The Proof: Runtime

Work for each candidate S: $O(n+m)$.

- Check that S is Independent Set and construct $G_{\setminus S}$.
- Run BFS to 2-color the remaining graph; construct the output if successful.

Number of candidates:

$$N(n) = \sum_{k=0}^{\lfloor n/3 \rfloor} \binom{n}{k}.$$

Thus the total time is

$$O((n+m)N(n)).$$

A General Combinatorial Bound

Lemma. For $0 < p \leq 1/2$,

$$\sum_{k=0}^{\lfloor pn \rfloor} \binom{n}{k} \leq \left(\left(\frac{1}{p} \right)^{p} \left(\frac{1}{1-p} \right)^{1-p} \right)^{n}.$$

A General Combinatorial Bound

Lemma. For $0 < p \le 1/2$,

$$\sum_{k=0}^{\lfloor pn \rfloor} \binom{n}{k} \le \left(\left(\frac{1}{p}\right)^{p} \left(\frac{1}{1-p}\right)^{1-p} \right)^{n}.$$

For $p = 1/3$, this gives

$$N(n) \le c^{n}, \qquad c = 3^{1/3} \left(\frac{3}{2}\right)^{2/3} \approx 1.8898816 < 1.89.$$

A General Combinatorial Bound

Lemma. For $0 < p \le 1/2$,

$$\sum_{k=0}^{\lfloor pn \rfloor} \binom{n}{k} \le \left(\left(\frac{1}{p}\right)^{p} \left(\frac{1}{1-p}\right)^{1-p} \right)^{n}.$$

For $p = 1/3$, this gives

$$N(n) \le c^{n}, \qquad c = 3^{1/3} \left(\frac{3}{2}\right)^{2/3} \approx 1.8898816 < 1.89.$$

Since $n + m = O(n^2)$ and $c < 1.89$, the polynomial factor is absorbed:

$$O\big((n+m)c^{n}\big) = O(1.89^{n}).$$